

# ABHAS KUMAR

Birla Institute of Technology Mesra • Ranchi, Jharkhand

[Email](#) • [Portfolio](#) • [GitHub](#) • [LinkedIn](#)

## EDUCATION

**Birla Institute of Technology Mesra, Ranchi** 2023 – 2028 (Expected)  
Integrated M.Sc. in Mathematics and Computing | Cumulative GPA: 7.4 / 10

## EXPERIENCE

**Management Trainee Intern — Wings Publication International** Dec 2024 – Feb 2025

Web Development & AWS Deployment • ₹12,000/month • Remote

- Built and deployed **multiple author landing pages** using WordPress and Elementor, managing full project lifecycle from design to go-live
- Configured production environment on **AWS EC2**, handling domain setup, SSL, and server management; received a **written testimonial from a Dubai-based client**

**Freelance ML Engineer — Dynode AI (Client: Zoran)** Mar 2026

Solar Energy Trading — AI Model Development • €800 EUR • Remote

- Ran a **5-fold TimeSeriesCV model comparison** across LightGBM, XGBoost, RandomForest, CatBoost, Ridge, and Stacking for PV production forecasting; **LightGBM selected** with MAE 0.12 ( $\pm 0.02$ ), RMSE 0.26
- Applied client's proprietary backtest over a **184-day simulation**, improving optimized profit from **€146K → €167K** and boosting improvement margin from **1.9% → 8.7%** over naive strategy

## PROJECTS

**TeachBridge — Full-Stack EdTech Platform** 2025

React • Node.js • Express • MongoDB • Live: [teach-bridge.vercel.app](https://teach-bridge.vercel.app) • [github.com/abhas-singh15/TeachBridge](https://github.com/abhas-singh15/TeachBridge)

- Built a **nonprofit-style tutoring platform** connecting students and tutors — features **user authentication, rolebased dashboards** (student & teacher), and a **booking system** handling multiple concurrent session requests
- React** frontend on **Vercel**, **Node.js/Express** REST API on **Render**, **MongoDB** data layer — clean REST architecture with environment-based config and independent frontend/backend deployment

**Short-Term Solar Energy Forecasting — ML Research** 2025 – 2026

Python • scikit-learn • statsmodels • LightGBM • Pandas • [github.com/abhas-singh15/renewable-forecast](https://github.com/abhas-singh15/renewable-forecast)

- Forecasted hourly PV output over a **7-day horizon** (60-day rolling window); benchmarked Persistence, SARIMA, Random Forest, and **MLP — best performance (MAE 887, RMSE 1419)**
- Applied **cyclical feature encoding** (sin/cos of hour) and meteorological features; delivered fully reproducible Jupyter notebooks and a **downloadable research report** on GitHub

## SKILLS

**Languages:** Python, JavaScript, SQL, HTML/CSS

**Frontend:** React, WordPress, Elementor

**Backend:** Node.js, Express.js, REST APIs

**ML / Data:** LightGBM, XGBoost, RandomForest, MLP, SARIMA, Scikit-learn, Pandas, NumPy, statsmodels

**Databases & Cloud:** MongoDB, AWS EC2, Vercel, Render

**Tools:** Git, GitHub, Jupyter Notebook